

NCPOR Dataset Metadata PostgreSQL Schema

Schema rebuilt from the supplied NCPOR NPDC dataset export

Source records analyzed: 900 JSON records and 900 CSV rows. The supplied records expose NCPOR NPDC metadata such as title, abstract, purpose, release information, scientific classification, project details, personnel, data center, instrument, platform, location and resource links.

Design principle: preserve the source metadata faithfully, normalize repeated entities for relational querying, and retain the original raw metadata in JSONB so fields that are sparse or added by NCPOR later are not lost.

1. What the supplied dataset actually contains

The export includes a source identifier, title, abstract, release date and URLs at the record level. A representative record also contains structured Basic Information, Science Keywords, Project Details, Summary, Dataset Citation, Scientist Details, Data Center, Instrument Details, Location, Platform and Data Set Progress. ■filecite■turn0file0■L3-L59■

Observed field	Purpose in schema	Observed coverage
source	Core dataset/resource metadata	900 populated / 900
source_id	Core dataset/resource metadata	900 populated / 900
title	Core dataset/resource metadata	900 populated / 900
abstract	Core dataset/resource metadata	900 populated / 900
release_date	Core dataset/resource metadata	900 populated / 900
detail_url	Core dataset/resource metadata	900 populated / 900
xml_url	Core dataset/resource metadata	900 populated / 900
pdf_url	Core dataset/resource metadata	900 populated / 900
download_url	Core dataset/resource metadata	467 populated / 900
original_resource_url	Core dataset/resource metadata	210 populated / 900
raw_metadata	Core dataset/resource metadata	900 populated / 900

2. Normalized database model

The dataset-specific model is intentionally centered on **datasets**. Personnel, science keywords, data centers, instruments, platforms, locations, external/download resources, versions and embeddings are separate child tables because these concepts repeat across records or require their own lifecycle. The source record itself is preserved through **raw_metadata JSONB**.

datasets

Core dataset identity, descriptive metadata, publication/release metadata, and source links.

Column	Type	Constraint / meaning
id	UUID	PK
npdc_metadata_id	VARCHAR(64)	UNIQUE, NOT NULL — e.g. metadata record ID
source	VARCHAR(150)	NOT NULL
source_url	TEXT	Nullable
title	TEXT	NOT NULL
abstract	TEXT	Nullable

Column	Type	Constraint / meaning
purpose	TEXT	Nullable
release_date	DATE	Nullable
data_set_progress	VARCHAR(50)	Nullable
data_set_language	VARCHAR(50)	Nullable
originating_center	VARCHAR(255)	Nullable
expedition_type	VARCHAR(30)	Nullable
expedition_year	VARCHAR(30)	Nullable
project_number	VARCHAR(100)	Nullable
project_name	TEXT	Nullable
series_name	VARCHAR(100)	Nullable
dataset_title	TEXT	Nullable
release_place	VARCHAR(255)	Nullable
version	VARCHAR(50)	Nullable
detail_url	TEXT	Nullable
xml_url	TEXT	Nullable
pdf_url	TEXT	Nullable
download_url	TEXT	Nullable
original_resource_url	TEXT	Nullable
raw_metadata	JSONB	Nullable — preserve source metadata
created_at	TIMESTAMPTZ	DEFAULT now()
updated_at	TIMESTAMPTZ	DEFAULT now()

dataset_science_keywords

Scientific classification and keyword metadata observed in the source.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
category	VARCHAR(255)	Nullable
topic	VARCHAR(255)	Nullable
iso_topic	VARCHAR(255)	Nullable
keyword_text	TEXT	Nullable

dataset_personnel

Principal investigator / scientist information associated with a dataset.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
role	VARCHAR(100)	Nullable
name	VARCHAR(255)	Nullable
email	VARCHAR(320)	Nullable

Column	Type	Constraint / meaning
phone	VARCHAR(100)	Nullable
mobile	VARCHAR(100)	Nullable
organization_id	UUID	FK → institutions.id, nullable
address	TEXT	Nullable
city	VARCHAR(150)	Nullable
state	VARCHAR(150)	Nullable
postal_code	VARCHAR(30)	Nullable
country	VARCHAR(100)	Nullable

dataset_data_centers

Data center responsible for or associated with the dataset.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
short_name	VARCHAR(100)	Nullable
long_name	VARCHAR(255)	Nullable
email	VARCHAR(320)	Nullable
phone	VARCHAR(100)	Nullable
address	TEXT	Nullable
city	VARCHAR(150)	Nullable
state	VARCHAR(150)	Nullable
pin_code	VARCHAR(30)	Nullable
country	VARCHAR(100)	Nullable

dataset_instruments

Instrument metadata. The supplied records often have empty instrument values, so all fields remain nullable.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
short_name	VARCHAR(150)	Nullable
long_name	VARCHAR(255)	Nullable
metadata	JSONB	Nullable

dataset_platforms

Platform metadata for acquisition/observation context.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
short_name	VARCHAR(150)	Nullable
long_name	VARCHAR(255)	Nullable

Column	Type	Constraint / meaning
metadata	JSONB	Nullable

dataset_locations

Polar location context from the NPDC metadata.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
category	VARCHAR(255)	Nullable
region	VARCHAR(100)	Nullable
location_type	VARCHAR(150)	Nullable
subregion	VARCHAR(255)	Nullable
latitude	DOUBLE PRECISION	Nullable
longitude	DOUBLE PRECISION	Nullable
geometry	JSONB	Nullable — future spatial extension

dataset_resources

Downloadable or externally hosted resources linked to a dataset.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
resource_type	VARCHAR(50)	Nullable — data/pdf/xml/original/etc.
label	VARCHAR(255)	Nullable
url	TEXT	NOT NULL
mime_type	VARCHAR(150)	Nullable
storage_key	TEXT	Nullable — for object storage
file_size_bytes	BIGINT	Nullable
checksum	VARCHAR(128)	Nullable
is_primary	BOOLEAN	DEFAULT false

dataset_versions

Version history for future re-releases of the same dataset.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
version_label	VARCHAR(50)	NOT NULL
release_date	DATE	Nullable
change_summary	TEXT	Nullable
resource_id	UUID	FK → dataset_resources.id, nullable
created_at	TIMESTAMPTZ	DEFAULT now()

dataset_embeddings

Semantic-search chunks generated from dataset metadata/content.

Column	Type	Constraint / meaning
id	UUID	PK
dataset_id	UUID	FK → datasets.id
chunk_index	INTEGER	NOT NULL
chunk_text	TEXT	NOT NULL
embedding	VECTOR(1536)	Model-dependent
model_name	VARCHAR(150)	NOT NULL
metadata	JSONB	Nullable
created_at	TIMESTAMPTZ	DEFAULT now()

3. PostgreSQL DDL

Enable required extensions:

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;  
CREATE EXTENSION IF NOT EXISTS vector;
```

Core table:

```
CREATE TABLE datasets (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  npdc_metadata_id VARCHAR(64) UNIQUE NOT NULL,  
  source VARCHAR(150) NOT NULL,  
  source_url TEXT,  
  title TEXT NOT NULL,  
  abstract TEXT,  
  purpose TEXT,  
  release_date DATE,  
  data_set_progress VARCHAR(50),  
  data_set_language VARCHAR(50),  
  originating_center VARCHAR(255),  
  expedition_type VARCHAR(30),  
  expedition_year VARCHAR(30),  
  project_number VARCHAR(100),  
  project_name TEXT,  
  series_name VARCHAR(100),  
  dataset_title TEXT,  
  release_place VARCHAR(255),  
  version VARCHAR(50),  
  detail_url TEXT,  
  xml_url TEXT,  
  pdf_url TEXT,  
  download_url TEXT,  
  original_resource_url TEXT,  
  raw_metadata JSONB,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

Science keywords:

```
CREATE TABLE dataset_science_keywords (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,  
  category VARCHAR(255),  
  topic VARCHAR(255),  
  iso_topic VARCHAR(255),  
  keyword_text TEXT  
);
```

Personnel:

```
CREATE TABLE dataset_personnel (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,  
  role VARCHAR(100),  
  name VARCHAR(255),  
  email VARCHAR(320),  
  phone VARCHAR(100),  
  mobile VARCHAR(100),  
  organization_id UUID REFERENCES institutions(id) ON DELETE SET NULL,  
  address TEXT,  
  city VARCHAR(150),  
  state VARCHAR(150),  
  postal_code VARCHAR(30),  
  country VARCHAR(100)  
);
```

Data center:

```
CREATE TABLE dataset_data_centers (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,  
  short_name VARCHAR(100),  
  long_name VARCHAR(255),  
  email VARCHAR(320),  
  phone VARCHAR(100),  
  address TEXT,  
  city VARCHAR(150),  
  state VARCHAR(150),  
  pin_code VARCHAR(30),  
  country VARCHAR(100)  
);
```

Instrument and platform:

```
CREATE TABLE dataset_instruments (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

        dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
        short_name VARCHAR(150),
        long_name VARCHAR(255),
        metadata JSONB
    );

CREATE TABLE dataset_platforms (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
    short_name VARCHAR(150),
    long_name VARCHAR(255),
    metadata JSONB
);

```

Location:

```

CREATE TABLE dataset_locations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
    category VARCHAR(255),
    region VARCHAR(100),
    location_type VARCHAR(150),
    subregion VARCHAR(255),
    latitude DOUBLE PRECISION,
    longitude DOUBLE PRECISION,
    geometry JSONB
);

```

Resources:

```

CREATE TABLE dataset_resources (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
    resource_type VARCHAR(50),
    label VARCHAR(255),
    url TEXT NOT NULL,
    mime_type VARCHAR(150),
    storage_key TEXT,
    file_size_bytes BIGINT,
    checksum VARCHAR(128),
    is_primary BOOLEAN NOT NULL DEFAULT false
);

```

Versions:

```

CREATE TABLE dataset_versions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
    version_label VARCHAR(50) NOT NULL,
    release_date DATE,
    change_summary TEXT,
    resource_id UUID REFERENCES dataset_resources(id) ON DELETE SET NULL,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

Embeddings:

```

CREATE TABLE dataset_embeddings (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    dataset_id UUID NOT NULL REFERENCES datasets(id) ON DELETE CASCADE,
    chunk_index INTEGER NOT NULL,
    chunk_text TEXT NOT NULL,
    embedding VECTOR(1536),
    model_name VARCHAR(150) NOT NULL,
    metadata JSONB,
    created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
    UNIQUE (dataset_id, chunk_index)
);

```

4. Indexing and semantic-search support

```
CREATE INDEX idx_datasets_release_date
ON datasets(release_date);

CREATE INDEX idx_datasets_expedition_type
ON datasets(expedition_type);

CREATE INDEX idx_datasets_project_number
ON datasets(project_number);

CREATE INDEX idx_dataset_keywords_category_topic
ON dataset_science_keywords(category, topic);

CREATE INDEX idx_dataset_locations_region_subregion
ON dataset_locations(region, subregion);

CREATE INDEX idx_dataset_resources_dataset
ON dataset_resources(dataset_id);

CREATE INDEX idx_dataset_versions_dataset
ON dataset_versions(dataset_id);

CREATE INDEX idx_dataset_embeddings_dataset
ON dataset_embeddings(dataset_id);

-- HNSW index; tune vector dimension/model to the selected embedding model.
CREATE INDEX idx_dataset_embeddings_vector
ON dataset_embeddings
USING hnsw (embedding vector_cosine_ops);
```

For pgvector, VECTOR(1536) is only an example dimension. The production dimension must match the embedding model actually selected by the team. If the embedding model changes, use a migration/versioning strategy rather than mixing incompatible vectors.

5. Mapping from NCPOR metadata to the new schema

NCPOR source section	Target table / columns
Record identity: id, source, source_id, title, abstract, release_date	datasets.npdc_metadata_id, source, title, abstract, release_date
Basic Information: progress, language, originating center	datasets.data_set_progress, data_set_language, originating_center
Science Keywords: category, topic, ISO topic, expedition year	dataset_science_keywords + datasets.expedition_year
Project Details: project number, project name	datasets.project_number, datasets.project_name
Dataset Citation: creator, editor, dataset title, series, release place, version	dataset_personnel / datasets.dataset_title, series_name, release_place, version
Scientist Details	dataset_personnel
Data Center	dataset_data_centers
Instrument Details	dataset_instruments
Location: type, subregion and future coordinates	dataset_locations
Platform	dataset_platforms
Online/download/PDF/XML/original resource	datasets URL columns + dataset_resources
Original structured metadata	datasets.raw_metadata JSONB

6. Source-data observations that affect the schema

The records are sparse: many top-level fields are null even though corresponding information appears in raw_metadata/raw_text. For example, one supplied record has null values for fields such as science_category, project_number and scientist_name at the top level, while the raw metadata contains the corresponding structured values.

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The source also exposes resource links including detail, XML and PDF exports, and in some records a download URL and original resource URL. These should be modeled as resources rather than assuming every dataset has one physical file.

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Location is currently represented primarily by Antarctic/Arctic type and subregion metadata. Latitude/longitude are therefore nullable in the proposed schema; they should be populated only when the source or a verified enrichment supplies coordinates.

7. Example seed mapping

```
INSERT INTO datasets (  
  npdc_metadata_id, source, source_url, title, abstract,  
  release_date, data_set_progress, expedition_type,  
  expedition_year, project_number, project_name, version  
) VALUES (  
  'MF-651581924',  
  'NCPOR NPDC',  
  'https://npdc.ncpor.res.in/npdc/search_mf_data.action?...',  
  'Atmospheric Studies using Moveable Atmospheric Radar for Antarctica',  
  'Radar observations of atmospheric structure and turbulence...',  
  DATE '2021-03-01',  
  'Complete',  
  'Antarctic',  
  '2017-2018',  
  'AMS-12',  
  'Atmospheric Studies using Moveable Atmospheric Radar for Antarctica',  
  '1'  
);
```

8. Recommended relationship to the wider portal schema

This dataset model plugs into the previously designed portal through `dataset_id` relationships from publications, research projects, expeditions, media and saved items. The dataset remains the authoritative metadata entity; binary files belong in object storage, while PostgreSQL stores metadata, permissions, versions, provenance and searchable representations.

Important implementation note: keep the original NCPOR metadata payload. This gives the team a safe fallback when new metadata fields appear and supports audit/provenance without redesigning the relational schema for every source-side change.